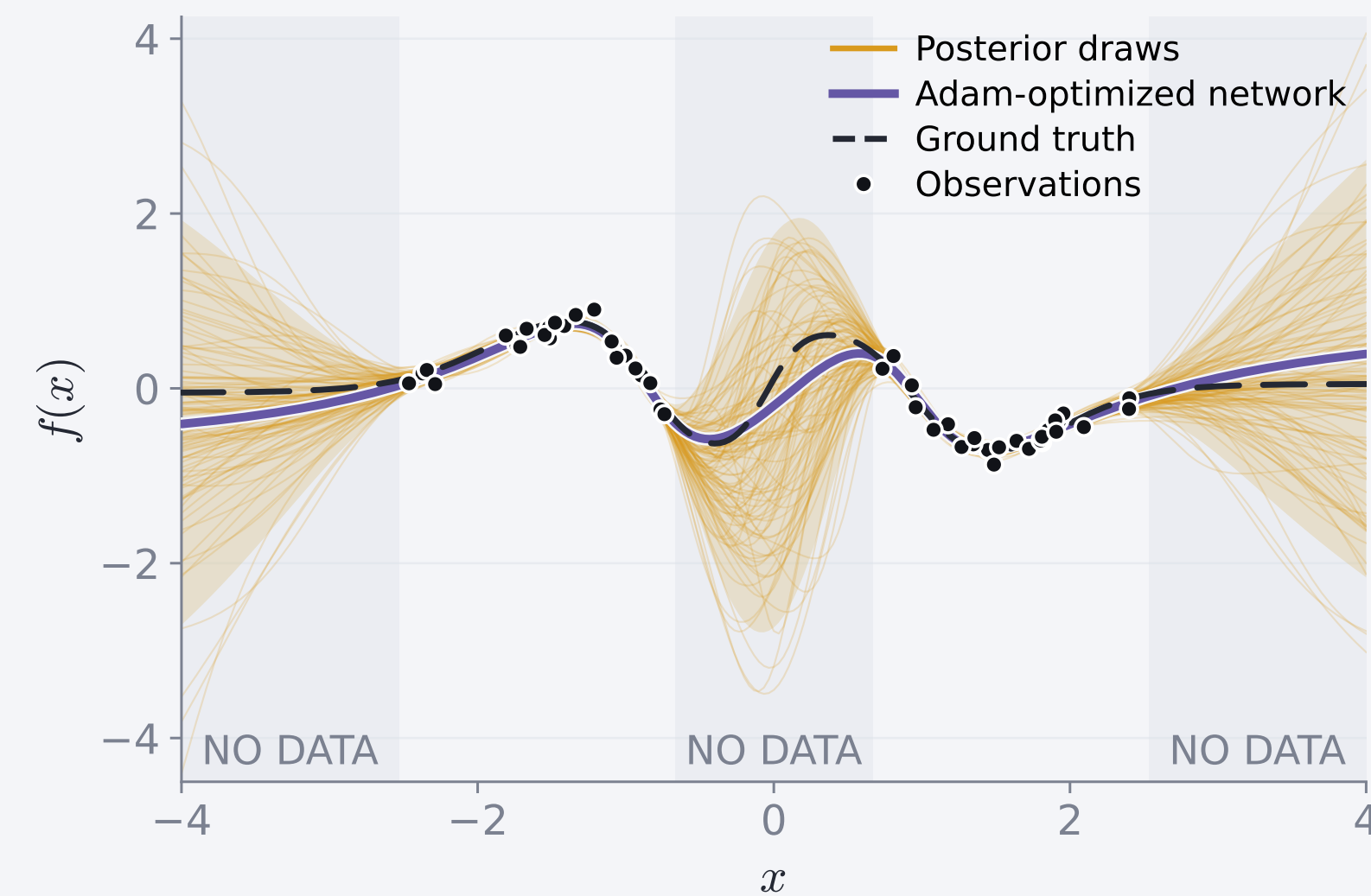




Why sample a neural network?

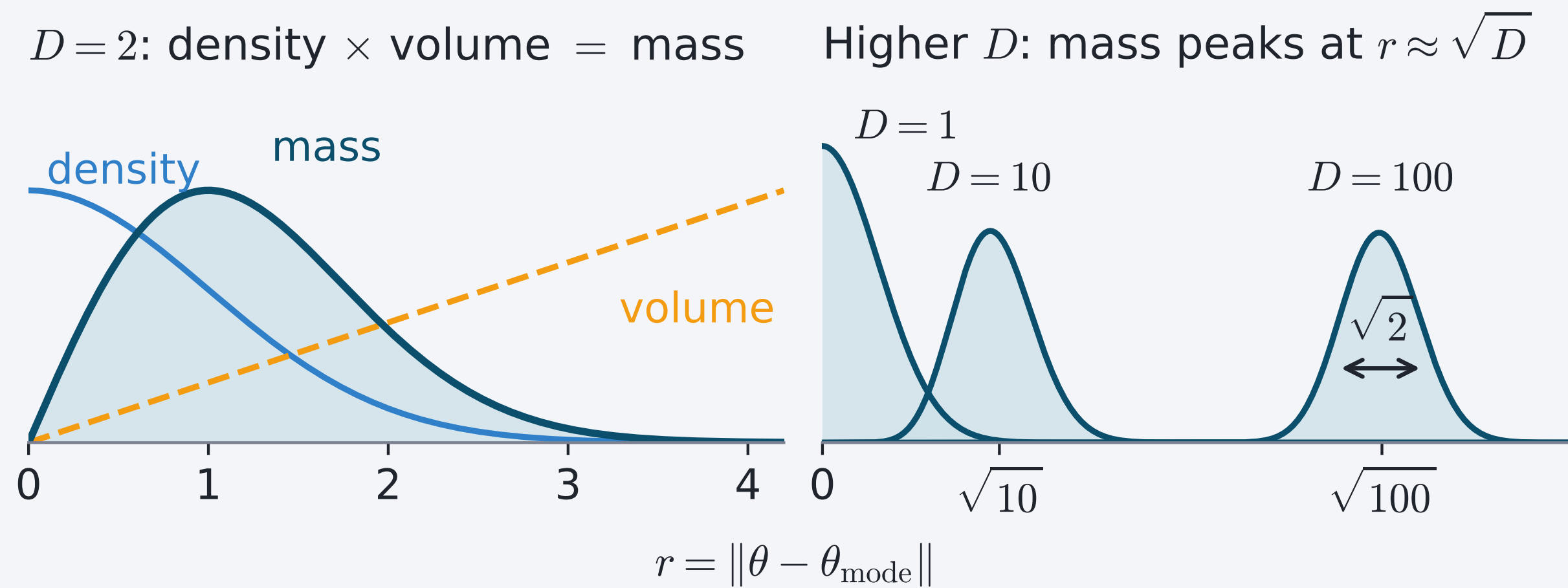


Ordinary training: one set of weights $\rightarrow$ one function.

Bayesian sampling: a distribution over weights $\theta \rightarrow$ uncertainty over functions.

$$\pi(\theta) = p(\theta | \mathcal{D}) \propto \underbrace{p(\mathcal{D} | \theta)}_{\text{fit}} \underbrace{p(\theta)}_{\text{prior}}$$

Where the probability lives

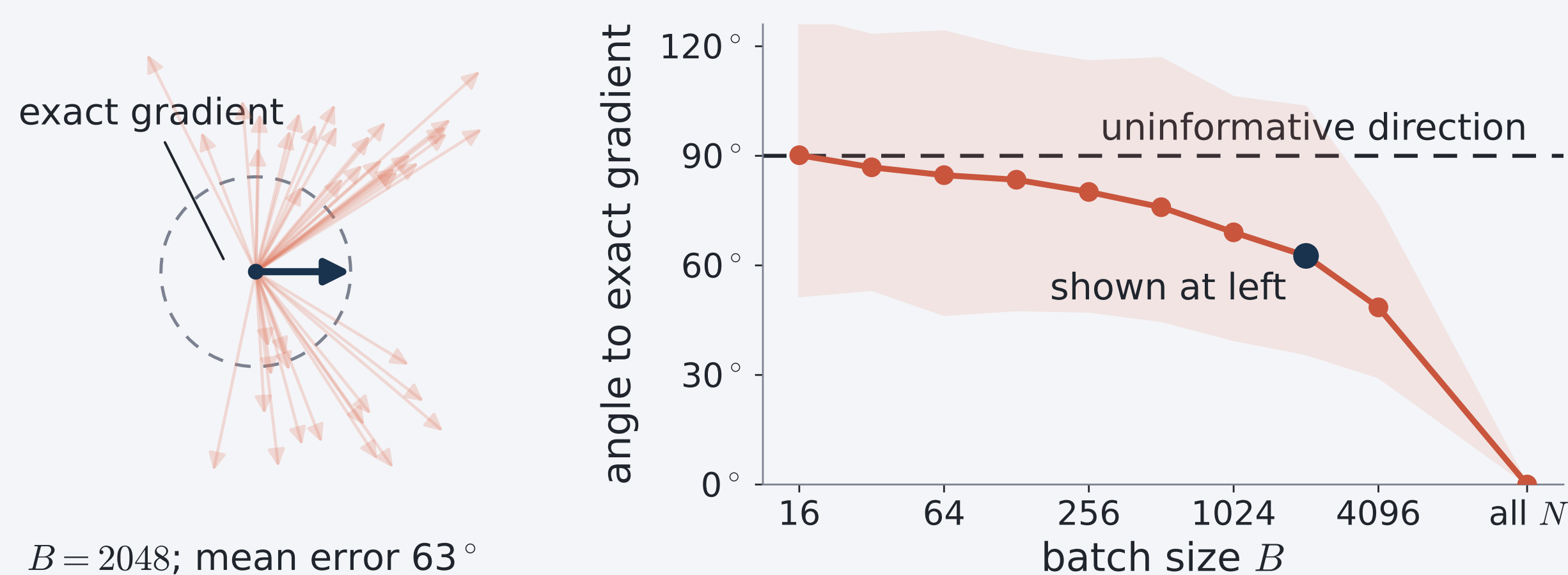


Because volume grows as r^{D-1} , probability concentrates in the *typical set*: a thin shell near $\|\theta\|^2 \approx D$. A correct sampler therefore spends almost all its time on this shell.

MCMC in the era of big data

Markov chain Monte Carlo (MCMC) generates a sequence of correlated samples targeting π . Gradient-based samplers use $\nabla \log \pi(\theta)$ to guide their updates. Computing this gradient exactly uses all N data points. A minibatch of size $B \ll N$ is unbiased and roughly N/B times cheaper, but never exact:

$$\hat{g}_B(\theta) = \nabla \log p(\theta) + \frac{1}{B} \sum_{i \in B} \nabla \log p(\mathcal{D}_i | \theta), \quad \text{Cov}[\hat{g}_B] \propto 1/B.$$



HMC and ray tracing under gradient noise

Hamiltonian Monte Carlo (HMC)^[1] and ray tracing (RT)^[2] are gradient-based samplers that both use $\nabla \log \pi$. HMC evolves an unconstrained velocity, whereas RT evolves a unit direction.

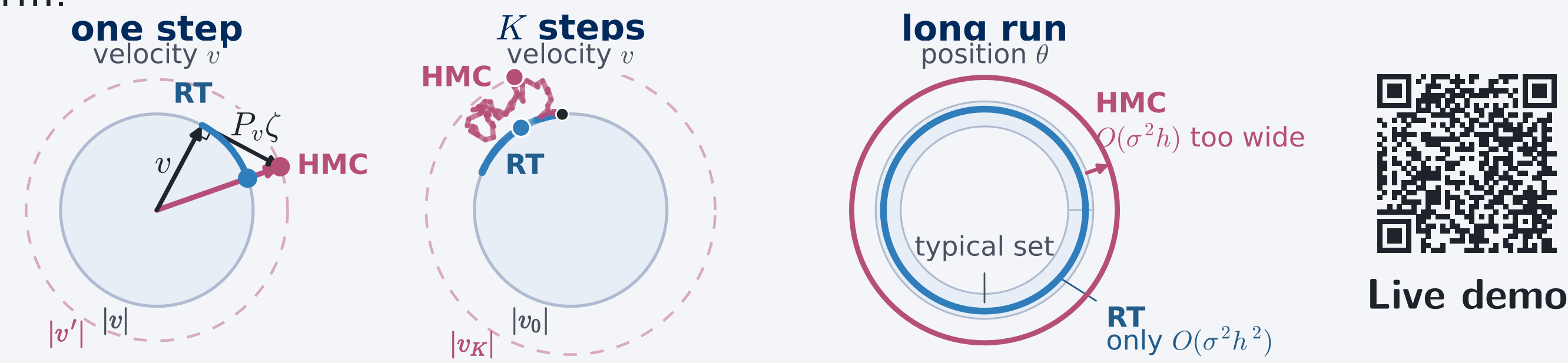
1. One noisy step

Without a Metropolis correction, mean-zero gradient noise raises HMC's expected kinetic energy, increasing its speed.

For RT, write the isotropic gradient error as $\sigma \zeta$. With $n(\theta) = \pi(\theta)^{1/(D-1)}$, the unit direction obeys

$$\frac{du}{ds} = P_u \left(\nabla \log n(\theta) + \frac{\sigma}{D-1} \zeta \right), \quad P_u = I - uu^\top.$$

P_u projects onto the tangent plane, so noise can rotate u but cannot change its norm.



2. Along a trajectory

With step size h and trajectory length S , the S/h noisy kicks accumulate as

$$\frac{S}{h} \cdot O(\sigma^2 h^2) = O(\sigma^2 h).$$

Let $\sigma_c(h)$ be the largest noise scale at step h that keeps long-run bias below a fixed tolerance, with $\sigma_c(h) \propto h^{-p}$. For unadjusted minibatch HMC, $p = \frac{1}{2}$.

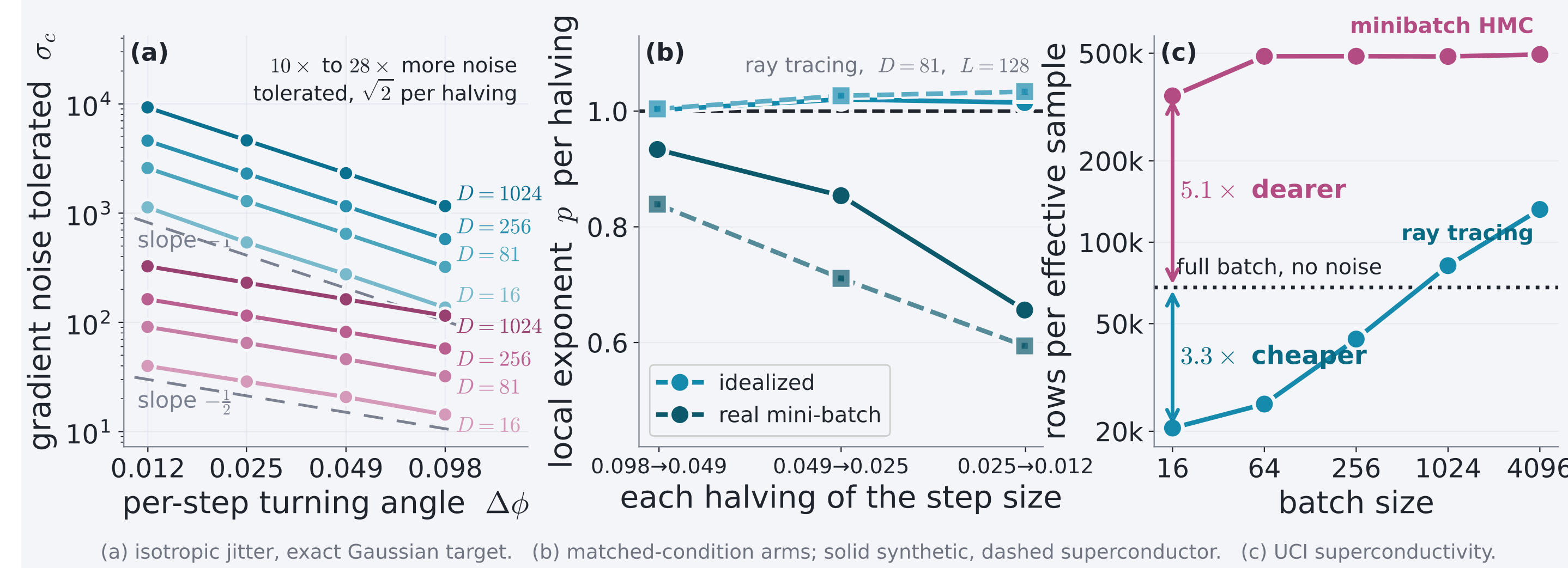
3. Long-run bias

For RT, isotropic rotations leave the uniform direction distribution unchanged, so the $O(\sigma^2 h)$ effect creates no leading bias:

$$b_{\text{RT}}(h, \sigma) = O(\sigma^2 h^2) + o(\sigma^2) \Rightarrow \sigma_c(h) = \Omega(h^{-1}), \quad p \geq 1$$

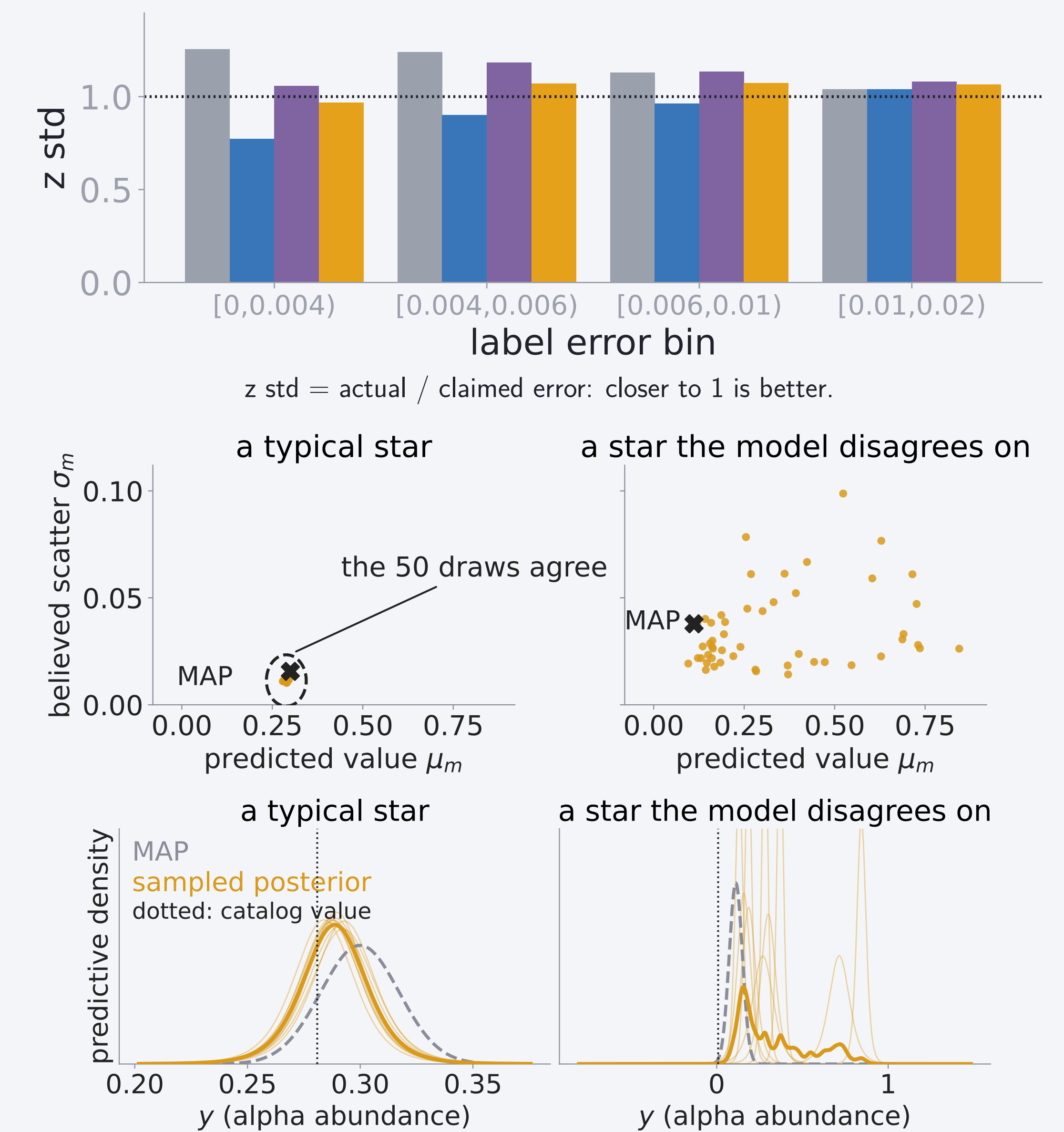
Measured noise tolerance and sampling cost

Across $D = 16$ –1024, idealized ray tracing fits $p = 1.00$; real minibatches lower the local exponent.



Uncertainty quality on the Gaia posterior

Result: only ray tracing keeps the error bars right in every group, and averaging its 50 draws makes each held-out star 27% more probable than the best single fit (MAP). Data^[4,5]: 126,156 giants, each a 110-coefficient Gaia XP spectrum with an APOGEE alpha label; a network ($D = 11,394$) maps spectrum to abundance and scatter, sampled with minibatch ray tracing. Key: **point (MAP)**, **deep ensemble**, **SGHMC**^[3], **ray tracing**.



Conclusion

- Noise cannot heat a ray that moves at constant speed, it can only turn it. That is why the tolerance exponent stays at $p = 1$ at every dimension we tested ($D = 16$ to 1024), while minibatch HMC sits at $p = \frac{1}{2}$.
- Real minibatches do break this clean picture and the exponent drops. Even so, the sampler keeps working and costs less: about $3.3\times$ less compute per effective sample, while minibatch HMC gets $5.1\times$ more expensive. Whether $p = 1$ holds exactly in stationarity is still open.
- On the Gaia data the method held up: the error bars come out right in every group of stars, and averaging the draws makes each held-out star 27% more probable than the single best fit. When the draws disagree about a star, that spread is a warning a single fit cannot give.

References

- [1] Betancourt, M. 2017, *A Conceptual Introduction to Hamiltonian Monte Carlo*, arXiv:1701.02434
- [2] Behroozi, P. 2026, *The Ray Tracing Sampler: Bayesian Sampling of Neural Networks for Everyone*, Open Journal of Astrophysics 9, doi:10.33232/001c.165932
- [3] Chen, T., Fox, E. B., Guestrin, C. 2014, *Stochastic Gradient Hamiltonian Monte Carlo*, ICML, PMLR 32(2), 1683
- [4] Gaia Collaboration, Vallenari, A., et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1
- [5] Laroche, A., Speagle, J. S. 2025, *Closing the Stellar Labels Gap: Stellar Label Independent Evidence for $[\alpha/M]$ Information in Gaia BP/RP Spectra*, ApJ 979, 5